CSC 251 HOMEWORK SIX

COMPUTING ELECTRIC POTENTIAL DUE TO 2 POINT CHARGES

THE ASSIGNMENT

You will use NumPy arrays and operations within a Jupyter notebook to compute electric potential around two point charges. You will use a NumPy meshgrid to assist in visualizing the electric potential in the region. Finally, you will determine the electric potential at a point in the region using your NumPy ndarrays.

ELECTRIC POTENTIAL

You may recall the concepts of an electric field and electric potential from your physics class. An electric field exits around any point charge. The strength of this field decreases by the square of the distance from the charge according to

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}$$

where 9 x 10^9 Newtons can be used for the $1/(4\pi\varepsilon_0)$ constant, q is the charge in Coulombs and magnitude of r is the distance from the point charge in meters.

The electric potential is defined as the amount of energy required to bring a second charged particle from infinity (a distance far away) to a distance r from the first point charge. The potential at each point in the field around the first point charge could be calculated according to

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$

in units of volts. If multiple point charges were present in the same region of space, the electric potential at a point in the field would be the sum of the electric potential due to the charges.

THE JUPYTER NOTEBOOK

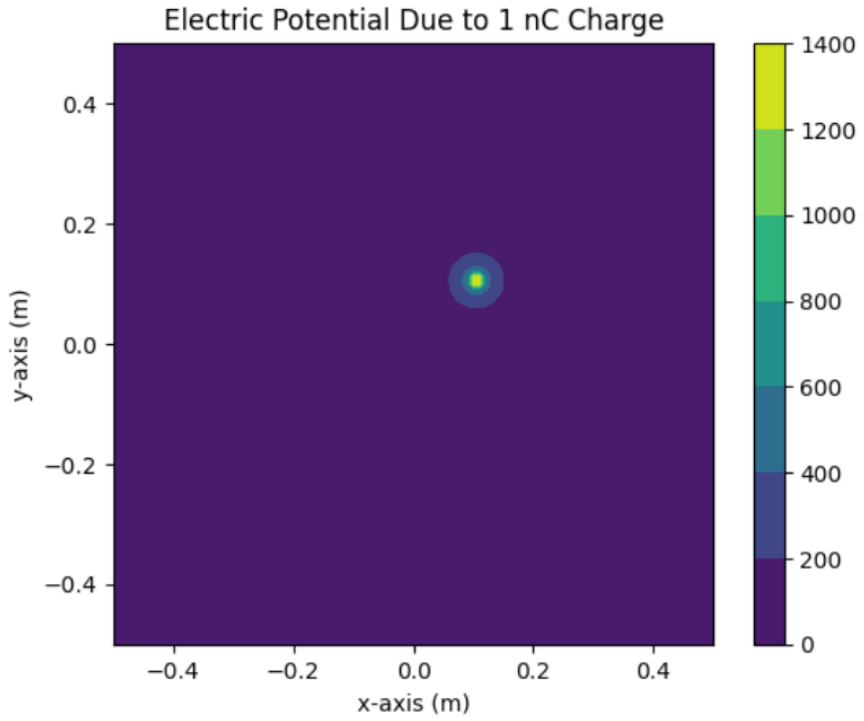
Follow these steps to complete this calculation in your notebook:

1) Import numpy and matplotlib.
2) Create two identical 1D ndarrays, x and y, to hold grid values from -.5 m up through .5 m (inclusive) with an interval of 0.01 m.
3) Create a NumPy meshgrid to create a grid of points using the x and y arrays.
4) Position particle 1 at (0.105, 0.105) with a charge of 1 nanocoulombs. Offsetting the particle from a grid point will prevent any divide by zero errors.
5) Compute the electric potential due to particle 1 for each point of the meshgrid.

6) Use a contour map to illustrate the electric potential values around charge 1.
7) Position particle 2 at (0.305, -0.305) with a charge of 2 nanocoulombs. Compute the electric potential due to particle 2 for each point of the meshgrid and create a contour map of the electric potential values.
8) Compute the total electric potential for all points in the region.
9) Use a contour map to show **both charges and the point (0,0)** as well as the electric potential values throughout the meshgrid.
10) Output the total electric potential at the point (0, 0) using code to output the value rather than visually picking the value out of the array.

SAMPLE OUTPUT

Your contour plot for the 1 nanocoulomb charge should look something like this:



NOTEBOOK REQUIREMENTS

Your notebook should contain a **markdown** cell at the top with your name, date, and assignment description. Each code cell should contain a comment to describe the script in that cell. You may break up your script in as many cells as you find helpful to work your way through the problem. Additional visualizations are encouraged, however, clean up an extraneous script that does not assist in solving the problem.

You may use any functions from NumPy and matplotlib but not from other packages. You should have no need to explicitly iterate through your vectors since NumPy does it for you.

You will not be penalized for magic numbers. However, you should follow best practices of assigning constants to variables in all caps and other values such as the charge magnitudes and positions to lowercase variables. These don't need to necessarily be at the top of your notebook, but in a cell preceding each calculation.

Your final value for the electric potential for point (0, 0) should be included in a markdown cell at the bottom of your notebook with proper units indicated.

Run your notebook then save it as *`HW6LastNameFirstInitial.`*`ipynb` and upload to Moodle.